Soal_3_Manual

Soal:

3. Tim marketing memiliki tiga himpunan pelanggan berdasarkan kanal akuisisi:

Definisi Himpunan Pelanggan

A = {N, N+2, N+4, N+7, N+10, N+13, N+15} □ pelanggan dari media sosial

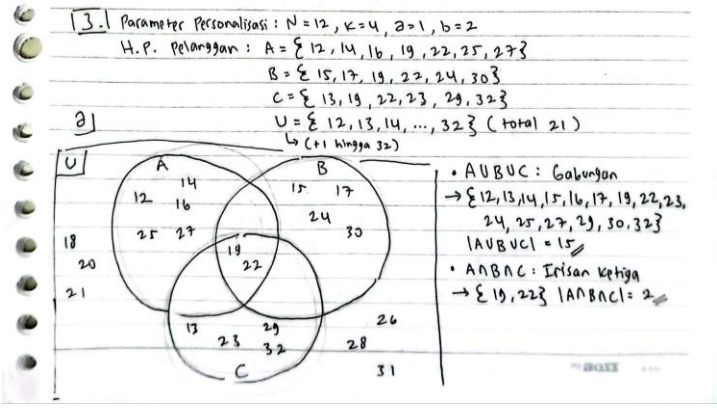
B = {N+3, N+5, N+7, N+10, N+12, N+18} □ pelanggan dari referral

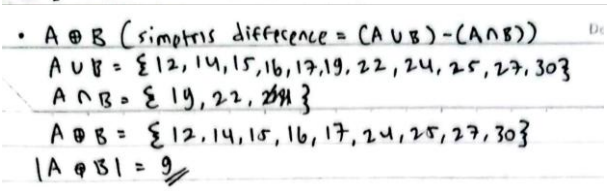
C = {N+1, N+7, N+10, N+11, N+17, N+20} □ pelanggan dari loyalty program

Universe U = {N, N+1, N+2, …, N+20} (total 21 pelanggan)

a) Gambar diagram Venn tiga himpunan ini secara manual (tulis tangan), lalu hitung manual: $|A \cup B \cup C|$, $|A \cap B \cap C|$, $|(A \cup B) \cap C'|$, dan $|A \oplus B|$ (simetris difference = $(A \cup B) - (A \cap B)$).
b) Verifikasi (a) dengan MATLAB menggunakan union, intersect, setdiff, dan setxor. **Screenshot output**.
c) Diberikan dua fungsi harga: $f(x) = K \cdot x + (a+b)$ (harga eceran) dan $g(x) = (x^2)/(b+1) + a$ (surcharge berdasarkan volume pembelian). Tentukan:
   - $(f \circ g)(x)$ dan $(g \circ f)(x)$, simplifikasikan sampai bentuk baku.
   - Apakah komposisi komutatif? Buktikan dengan contoh numerik x = b+2.
d) Tentukan $f^{-1}(x)$ secara manual. Tim akuntansi ingin tahu: jika pendapatan total dari satu pelanggan adalah Rp $(K \cdot 100 + a + b)$ ribu, berapa volume pembelian x yang menghasilkannya? Verifikasi hasil dengan subs dan solve di MATLAB.
e) Klasifikasikan fungsi f dan g: injektif? surjektif? bijektif? dengan domain dan kodomain $\mathbb{R}$. Justifikasi jawaban Anda dengan argumen matematis, bukan sekedar klaim.

Jawaban:





Kode soal 3b

```
% ====================================================
%  UTS Matematika Sains Data — Semester Genap 2025/2026
%  Soal 3 — Himpunan Fungsi
%  ---------------------------------------------------
%  Nama  : Ziohamdan Putra Yudhana
%  NIM   : 2510514012
%  Parameter: N=12, a=1, b=2, K=4, theta0=40, alpha=0.003
%  Tanggal: <2026-04-28>
% ====================================================

clc; clear; close all;

% Parameter personalisasi
N = 12;  a = 1;  b = 2;  K = 4;
theta0 = 40;  % derajat
alpha_lr = 0.003;

% Definisi himpunan
A = [N, N+2, N+4, N+7, N+10, N+13, N+15];
B = [N+3, N+5, N+7, N+10, N+12, N+18];
C = [N+1, N+7, N+10, N+11, N+17, N+20];
U = N : N+20;

fprintf('============================================\n');
fprintf('SOAL 3b: VERIFIKASI OPERASI HIMPUNAN\n');
fprintf('NIM: 2510514012\n');
fprintf('============================================\n\n');

fprintf('A = '); disp(A);
fprintf('B = '); disp(B);
fprintf('C = '); disp(C);
fprintf('U = '); disp(U);
fprintf('\n');

% Operasi himpunan
union_AB = union(A, B);
union_ABC = union(union(A, B), C);
intersect_ABC = intersect(intersect(A, B), C);

% (A ∪ B) ∩ C'
C_complement = setdiff(U, C);
union_AB_intersect_Ccomplement = intersect(union_AB, C_complement);

% Symmetric difference A ⊕ B
symdiff_AB = setxor(A, B);

fprintf('=== HASIL OPERASI HIMPUNAN ===\n');
fprintf('|A ∪ B ∪ C| = %d\n', length(union_ABC));
fprintf('Isi A ∪ B ∪ C: ');
fprintf('%.0f ', union_ABC);
fprintf('\n\n');

fprintf('|A ∩ B ∩ C| = %d\n', length(intersect_ABC));
fprintf('Isi A ∩ B ∩ C: ');
fprintf('%.0f ', intersect_ABC);
fprintf('\n\n');

fprintf('|(A ∪ B) ∩ C'| = %d\n', length(union_AB_intersect_Ccomplement));
fprintf('Isi (A ∪ B) ∩ C': ');
fprintf('%.0f ', union_AB_intersect_Ccomplement);
fprintf('\n\n');

fprintf('|A ⊕ B| = %d\n', length(symdiff_AB));
fprintf('Isi A ⊕ B (symmetric difference): ');
fprintf('%.0f ', symdiff_AB);
fprintf('\n\n');

fprintf('Timestamp: %s\n', datetime('now'));
```

Output:

```
==========================================
SOAL 3b: VERIFIKASI OPERASI HIMPUNAN
NIM: 2510514012
==========================================

A =     12    14    16    19    22    25    27

B =     15    17    19    22    24    30

C =     13    19    22    23    29    32

U =     12    13    14    15    16    17    18    19    20    21    22    23    24    25    26    27    28    29    30    31    32


=== HASIL OPERASI HIMPUNAN ===
|A ∪ B ∪ C| = 15
Isi A ∪ B ∪ C: 12 13 14 15 16 17 19 22 23 24 25 27 29 30 32

|A ∩ B ∩ C| = 2
Isi A ∩ B ∩ C: 19 22

|(A ∪ B) ∩ C'| = 9
Isi (A ∪ B) ∩ C': 12 14 15 16 17 24 25 27 30

|A ⊕ B| = 9
Isi A ⊕ B (symmetric difference): 12 14 15 16 17 24 25 27 30

Timestamp: 28-Apr-2026 17:16:11
```

c) • $f(x) = K \cdot x + (a+b) = 4x + 3$

• $g(x) = \frac{x^2}{(b+1)+a} = \frac{x^2}{3+1} + 1$

$(f \circ g)(x) = f(g(x)) = \cancel{4\left(\left(\frac{x^2}{3+1}\right)\right)+3} = 4\left(\frac{x^2}{3}+1\right)+3$

$= \frac{4x^2}{3} + 4 + 3 = \frac{4x^2}{3} + 7$

$(g \circ f)(x) = g(f(x)) = \frac{(4x+3)^2}{3} + 1 = \frac{(16x^2+24x+9)}{3} + 1$

$= \frac{16x^2}{3} + 8x + 3 + 1 = \frac{16x^2}{3} + 8x + 4$

Komposisi tidak komutatif karena $(f \circ g)(x) \neq (g \circ f)(x)$

untuk $x \neq 0$

contoh $x = 4$

$(f \circ g)(4) = \frac{4 \cdot (16)}{3} + 7 = \frac{64}{3} + 7 = 28{,}33$

$(g \circ f)(4) = \frac{16(16)}{3} + 8(4) + 4 = \frac{256}{3} + 32 + 4 = 121{,}33$

$28{,}33 \neq 121{,}33$, Tidak sama berarti tidak komutatif.

d) $f(x) = 4x + 3 \rightarrow$ misal $y = 4x + 3 \rightarrow x = \frac{y-3}{4}$

$f^{-1}(x) = \frac{(x-3)}{4}$

Jika pendapatan $= K \cdot 100 + a + b = 4 \cdot 100 + 3 = 403$ ribu

$f^{-1}(403) = \frac{(403-3)}{4} = 400/4 = 100$

Volume pembelian $x = 100$ unit

Kode soal 3d (verifikasi)

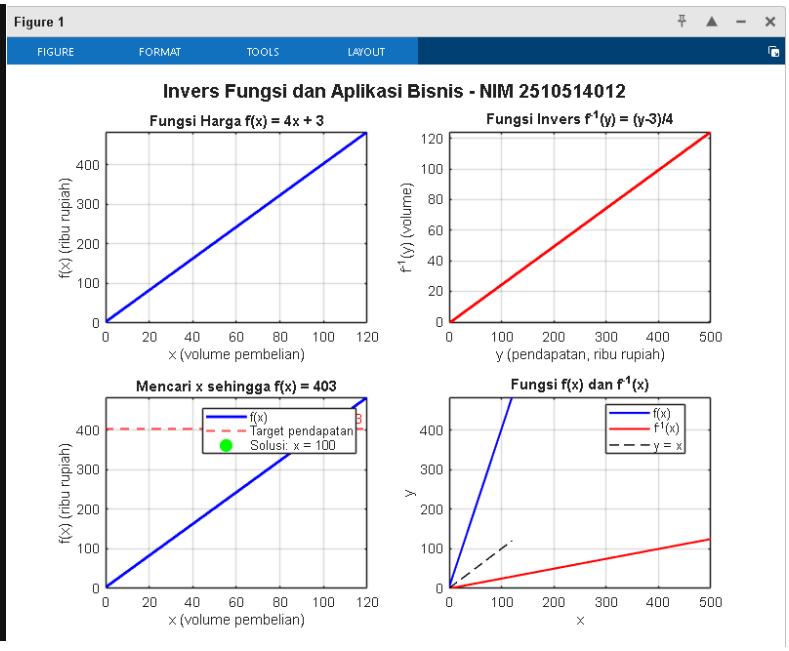
```
% ======================================================
%  UTS Matematika Sains Data — Semester Genap 2025/2026
%  Soal 3 — Himpunan Fungsi
%  ----------------------------------------------------
%  Nama  : Ziohamdan Putra Yudhana
%  NIM   : 2510514012
%  Parameter: N=12, a=1, b=2, K=4, theta0=40, alpha=0.003
%  Tanggal: <2026-04-28>
% ======================================================

clc; clear; close all;

% Parameter personalisasi
N = 12;  a = 1;  b = 2;  K = 4;
theta0 = 40;  % derajat
alpha_lr = 0.003;
```

```
% Definisi fungsi
syms x
f = K*x + (a+b);  % f(x) = 4x + 3
g = x^2/(b+1) + a; % g(x) = x^2/3 + 1

fprintf('==========================================\n');
fprintf('SOAL 3d: INVERS FUNGSI DAN VERIFIKASI\n');
fprintf('NIM: 2510514012\n');
fprintf('==========================================\n\n');

fprintf('f(x) = %s\n', char(f));
fprintf('g(x) = %s\n\n', char(g));

%% Mencari invers f⁻¹(x)
% f(x) = 4x + 3 = y
% 4x = y - 3
% x = (y - 3)/4
% f⁻¹(y) = (y - 3)/4

f_inv = (x - 3)/4;
fprintf('f⁻¹(x) = %s\n\n', char(f_inv));

% Verifikasi f(f⁻¹(x)) = x dan f⁻¹(f(x)) = x
f_compose_inv = subs(f, x, f_inv);
f_inv_compose_f = subs(f_inv, x, f);

fprintf('=== VERIFIKASI INVERS ===\n');
fprintf('f(f⁻¹(x)) = %s\n', simplify(f_compose_inv));
fprintf('f⁻¹(f(x)) = %s\n\n', simplify(f_inv_compose_f));

%% Tim akuntansi: pendapatan total = (K·100 + a + b) ribu rupiah
pendapatan = K*100 + (a+b); % = 400 + 3 = 403 ribu rupiah
fprintf('=== APLIKASI BISNIS ===\n');
fprintf('Pendapatan total dari satu pelanggan = Rp %d.000\n', pendapatan);
fprintf('Pendapatan = f(x) = 4x + 3 = %d\n\n', pendapatan);

% Mencari volume pembelian x menggunakan solve
persamaan = f == pendapatan;
x_volume = solve(persamaan, x);

fprintf('Mencari x sehingga f(x) = %d:\n', pendapatan);
fprintf('4x + 3 = %d\n', pendapatan);
fprintf('4x = %d\n', pendapatan - 3);
fprintf('x = %d/4 = %d\n\n', pendapatan - 3, x_volume);

fprintf('Volume pembelian x = %.0f unit\n', double(x_volume));

% Verifikasi dengan subs
f_verif = subs(f, x, x_volume);
fprintf('\nVerifikasi dengan subs: f(%.0f) = %.0f = pendapatan target ✓\n', ...
    double(x_volume), double(f_verif));

%% Verifikasi dengan solve dan subs
fprintf('\n=== VERIFIKASI MENGGUNAKAN solve DAN subs ===\n');

% Menggunakan solve untuk mencari x dari f(x) = pendapatan
x_solve = solve(f == pendapatan, x);
fprintf('solve(f(x) = %d): x = %s\n', pendapatan, char(x_solve));

% Menggunakan subs untuk verifikasi
f_subs = subs(f, x, x_solve);
fprintf('subs(f, x, x_solve) = %s = pendapatan ✓\n', char(f_subs));

% Menggunakan f⁻¹ langsung
x_inv = subs(f_inv, x, pendapatan);
fprintf('f⁻¹(pendapatan) = (%d - 3)/4 = %.0f ✓\n', pendapatan, double(x_inv));

%% Visualisasi fungsi f(x) dan f⁻¹(x)
figure('Position', [100 100 800 600], 'Color', 'w');

% Plot f(x)
subplot(2,2,1);
fplot(f, [0, 120], 'b-', 'LineWidth', 2);
xlabel('x (volume pembelian)'); ylabel('f(x) (ribu rupiah)');
```

```
title('Fungsi Harga f(x) = 4x + 3');
grid on;

% Plot f⁻¹(x)
subplot(2,2,2);
fplot(f_inv, [0, 500], 'r-', 'LineWidth', 2);
xlabel('y (pendapatan, ribu rupiah)'); ylabel('f^{-1}(y) (volume)');
title('Fungsi Invers f^{-1}(y) = (y-3)/4');
grid on;

% Plot f(x) dan garis y = pendapatan
subplot(2,2,3);
fplot(f, [0, 120], 'b-', 'LineWidth', 2); hold on;
yline(pendapatan, 'r--', sprintf('Pendapatan = %d', pendapatan), 'LineWidth', 1.5);
x_volume_double = double(x_volume);
plot(x_volume_double, pendapatan, 'go', 'MarkerSize', 10, 'MarkerFaceColor', 'g');
xlabel('x (volume pembelian)'); ylabel('f(x) (ribu rupiah)');
title(sprintf('Mencari x sehingga f(x) = %d', pendapatan));
legend('f(x)', 'Target pendapatan', sprintf('Solusi: x = %.0f', x_volume_double));
grid on;

% Plot f(x) dan garis identitas
subplot(2,2,4);
fplot(f, [0, 120], 'b-', 'LineWidth', 1.5); hold on;
fplot(f_inv, [0, 500], 'r-', 'LineWidth', 1.5);
fplot(x, [0, 120], 'k--', 'LineWidth', 1);
xlabel('x'); ylabel('y');
title('Fungsi f(x) dan f^{-1}(x)');
legend('f(x)', 'f^{-1}(x)', 'y = x', 'Location', 'best');
grid on;

sgtitle(sprintf('Invers Fungsi dan Aplikasi Bisnis - NIM 2510514012'), ...
    'FontSize', 14, 'FontWeight', 'bold');

% Simpan figure
saveas(gcf, 'Soal3d_Invers_Fungsi_NIM_2510514012.png');
fprintf('\nVisualisasi telah disimpan sebagai: Soal3d_Invers_Fungsi_NIM_2510514012.png\n');
```

Output:

```
==========================================
SOAL 3d: INVERS FUNGSI DAN VERIFIKASI
NIM: 2510514012
==========================================

f(x) = 4*x + 3
g(x) = x^2/3 + 1

f⁻¹(x) = x/4 - 3/4

=== VERIFIKASI INVERS ===
f(f⁻¹(x)) = x
f⁻¹(f(x)) = x

=== APLIKASI BISNIS ===
Pendapatan total dari satu pelanggan = Rp 403.000
Pendapatan = f(x) = 4x + 3 = 403

Mencari x sehingga f(x) = 403:
4x + 3 = 403
4x = 400
x = 400/4 = 100

Volume pembelian x = 100 unit

Verifikasi dengan subs: f(100) = 403 = pendapatan target ✓

=== VERIFIKASI MENGGUNAKAN solve DAN subs ===
solve(f(x) = 403): x = 100
subs(f, x, x_solve) = 403 = pendapatan ✓
f⁻¹(pendapatan) = (403 - 3)/4 = 100 ✓
```



e) $f(x) = 4x + 3$:

- Injektif = Iya karena $f(x_1) = f(x_2) \rightarrow 4x_1 + 3 = 4x_2 + 3 \rightarrow x_1 = x_2$
- surjektif = Iya karena untuk setiap $y \in R$, ada $x = \frac{(y-3)}{4} \in R$
- Bijektif = Iya karena Injektif & surjektif

$f(x) = \frac{x^2}{3} + 1$:

- Injektif = Tidak karena $g(x) = g(-x)$, contoh $g(2) = g(-2) = \frac{7}{3}$
- surjektif = Tidak karena minimum $g(x) = 1$, sehingga untuk $y < 1$ tidak ada x
- Bijektif = Tidak karena tidak Injektif & surjektif